Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Fuller, Charles A. Ph.D.		
Project Title:	Primate Circadian Rhythms in the Martian Environment		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	95616-5270	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	02/01/2001	**End Date:**	11/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Robinson, Edward (University of California Davis) Hoban-Higgins, Tana (University of California Davis)		
Grant/Contract No.:	HPF.002.05		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	To maintain health and homeostasis, an organism must regulate each of its physiological systems in concert with all of the others and with the external environment. The Circadian Timing System (CTS) has evolved to allow such coordination of an organism?s physiology and behavior both internally and with the external 24.0 hr earth day. The mammalian CTS is adapted to the lighting environment found here on Earth. As we move toward exploration-class space missions, we will be exposing astronauts to a non-Earth lighting environments for increasing lengths of time. Such changes may include altered gravity and spectral, intensity and day-length differences. This raises the concern of whether or not humans will be able to synchronize to such an alien environment. For example, a Mars-type exploration would entail stays on Mars of one to two years. Compared with the Earth, the Martian environment has a photic spectrum shifted to the red, low illumination level, a periodicity of 24.63 hr, and a 0.38G gravitational field. The mammalian CTS is most sensitive to light of the blue-green wavelengths and adapted to synchronize to a 24.0 hr day. In		

	addition, light must be relatively bright to affect the CTS of primates, especially humans. Further, altered circadian rhythms and sensitivity to light, including CTS period, have been reported in both the microgravity environment of spaceflight and in hyperdynamic fields on the earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This program will examine the ability of the primate (male and female rhesus monkey) CTS to cope with the Martian environment. Included in our project are an examination of the responses of the rhesus CTS to the Martian day. Specifically, in order to determine if rhesus can entrain to the Martian day, we will characterize the long-term physiological and behavioral responses to the Martian photic environment. The Martian photic environment differs from that of Earth in both period and spectrum. In order to separate out period from spectral effects, we will perform an equivalent study using a Martian day-length combined with an earth light spectrum. We will use the primate model to initiate the development of countermeasures to assure optimum entrainment of the CTS. One of the countermeasures we will test are timed bright light pulses. In order to make predictions about the effect of exposure to the Martian gravitational field, we will determine the effects of exposure to gravitational fields of 1.0, 1.5 and 2.0G on the period of the circadian pacemaker. We will use these results to develop a G vs. period model to predict the effect the 0.38G Martian environment will have on the period of the circadian pacemaker. This model will be further used to develop testable countermeasures. Thus, this program will develop a primate model to evaluate physiological and behavioral consequences of long-term exposure of male and female subjects to the altered lighting and/or gravitational environments that will be encountered by astronauts on long duration missions and to test countermeasures for such environments.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 12/08/2025)